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Engr 095 FYRE Program Pathway 1: Sensing the World
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Reflection 1

So far in this module I have learned a lot through both lectures and hands on seeing and learning from different departments and people. The first labs we were given multimeters and pencils and paper to try and make and own a circuit to conduct electricity from a short lecture. In each lab we also have a worksheet to fill out that helps us stay on point and keep track of the different key words and concepts we are being exposed to. Then we were taken to Wilbur Powerhouse Makerspace and learned about the different machines and tools and how we can get trained to be able to use them. We used a vinyl cutting machine to make little masks for electrodes we were going to make to be able to make a fully working sensor. We each got one and were shown how the machine worked. Then we would go to Whittaker to learn more about material science and experiment with multimeters and cutting more special vinyl to see the difference between difference shape and electric resistance. The clean room and the MOCVD where we were guided around by professors and got a hands-on look by having to get into full PPE and really feel what it's like to be a scientist. We also practiced finding resistance in a metal. We then actually hands on used different metal liquids like Ag paste to paint and spin coat on to the masks on the vinyl to be used. We then were able to dry and use these electrodes to make a very minimal sensor that could sense dampness in the air based on the reading on a multimeter. Making our own little working butterfly LED circuits has also been a highlight for me, a real experiment that helped me learn about circuits through a straightforward assignment that we actually got to keep. We also have just started working on the programming and computer engineering side of things with learning a little bit of python and using a micro controller.

Throughout this whole process I have had a really rewarding and revealing experience. I have learned so much about how materials are made and so many processes that I have never even thought about. Like how much detail and science has gone behind the perfecting of phone chips and sensors we overlook every day. Also how many machines are automated to do honestly mind blowing (to me now at least) mini experiments for production and testing everyday. I have learned about so many more engineering processes and new ways of thinking and optimization that are happening in the world right now through hands-on experience through guided experiments with real materials and authentic results. I have been exposed to a great deal of high learning topics and machines through this module so far that has greatly opened my eyes to all this program and this school has to offer, it has forced me to think about a lot of little things and machines that I overlook and simplify in my head everyday.